

Computer Communication and Networks (COMN)

INFR10074

2025-26, Semester 2

<https://opencourse.inf.ed.ac.uk/comn>

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THE UNIVERSITY of EDINBURGH
informatics

icsa

Institute for Computing
Systems Architecture

Overview

- Comprehensive introductory course on computer networks
 - Hands-on course with emphasis on practical networking aspects
 - Assumes experience programming in Python as well as general computer systems background
 - No pre-requisite courses although recommended to take other related courses like operating systems
 - Helpful or essential to have taken this course if you are planning on doing a networking-oriented Honours or MSc project

Learning outcome

On completion of this course, the student will be able to:

- explain key networking concepts, principles, design issues and techniques at all protocol layers
- contrast between different types of networks (e.g., wide area networks vs. local area networks, wired vs. wireless) in terms of their characteristics and protocols used
- describe different types of networked applications and what underlying network protocols are needed to meet their diverse requirements
- distinguish between control and data planes in computer networks, and their corresponding architectures in real-world networks (including the Internet)
- understand reliable transport protocols and networked system architectures via implementation using Socket APIs, measurement and analysis

Teaching staff

- Course lecturers
 - Michio Honda (michio.honda@ed.ac.uk) – Course Organizer
 - Mahesh Marina (mahesh@ed.ac.uk)
- Teaching assistants (TAs)
 - Andrew Ferguson (Andrew.E.Ferguson@ed.ac.uk)
 - Tianyi Gao (tianyi.gao@ed.ac.uk)

Course activities

- On-campus lectures twice a week
- Weekly online office hours or tutorials over Teams to cover essential background material for assignments
- 5 biweekly quizzes starting from week 3 but not assessed
- 2 practical programming-based assignments but only 1 assessed
- Exam
- Piazza for asynchronous Q&A
- Course related announcements on the [course mailing list](#)

Syllabus

- Introduction to Computer Networks, the Internet, TCP/IP architecture and protocols
 - Key concepts: protocol, network architecture, reference models, layering, service, interface, multiplexing, switching and standards
 - Performance metrics
 - Physical layer overview
- Application Layer
 - Representative Internet applications (Web/HTTP, P2P, multimedia, etc.)
 - Developing network applications using socket API
- Transport Layer
 - Connectionless vs. connection-oriented communications
 - Reliable data transfer and congestion/flow/error control
 - UDP and TCP protocols and more

Syllabus (contd.)

- Network Layer
 - Control and data planes: routing vs. forwarding
 - IP protocol and addressing
 - Routing algorithms
- Link Layer and Local Area Networks (LANs)
 - Link layer services and implementation
 - Ethernet and switched LANs
- Advanced topics
 - Host Networking
 - Programmable Networks
 - Datacenter Networking
 - Modern Internet

Assessment

- Practical assignment (50%)
 1. Analysis of network traffic traces (0%)
 2. TBA (50%)
 - Don't ask me for an extension; I cannot grant any.
 - The correct place is the [ITO](#)
 - See the [policy on late coursework submission](#) first.
- Exam (50%)
 - Based on lectures, textbook and references (e.g., sections in secondary textbooks, research papers etc) provided in the lectures or [course schedule](#)
 - Contents covered in courseworks, including formative ones, are also examinable

Reading material

- Lectures cannot go into every detail; you will need to read textbooks and other resources referenced in lectures or [course schedule](#)
- Main text
 - J. F. Kurose and K. W. Ross, “Computer Networking: A Top-Down Approach” 8th edition, Pearson Education, 2021.
- Secondary texts
 - L. L. Peterson and B. S. Davie, “Computer Networks: A Systems Approach,” 6th edition, Morgan Kaufmann, 2020.
 - A. S. Tanenbaum, N. Feamster and D. J. Wetherall, “Computer Networks,” 6th edition, Pearson Education, 2021.